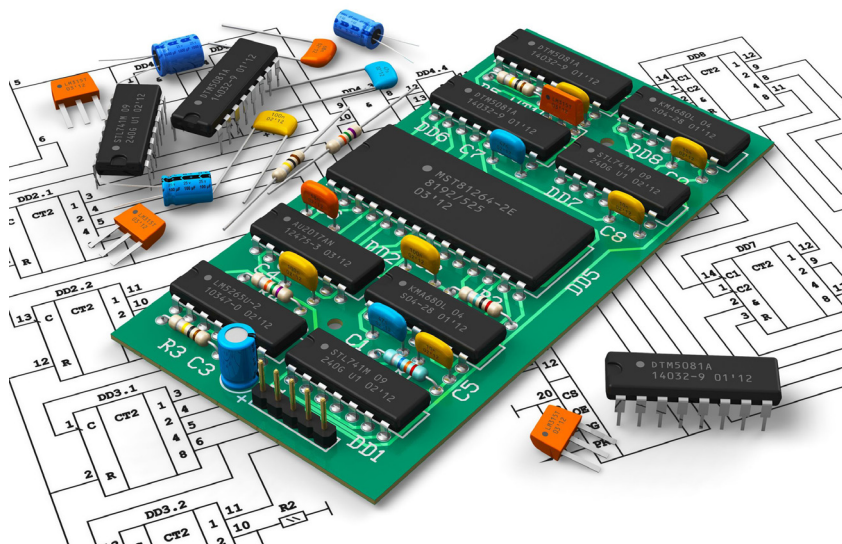




Components Are Key For Designers

An electronic product comprises components and, nowadays, some software as well. But components still form a major part of the product's cost in most cases. What if the components fail early and thus adversely affect the reputation of a brand? Not only will this bring a disrepute to the company, it may also affect their sales and their future. So, how do we ensure that this does not happen? Read on to find out



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The most important aspect of an electronic product is its components. A product's success or failure depends on the components used in the product. Normally, components form roughly 75% of the product cost. But improperly selected components can destroy the product's performance as well as damage the company's reputation beyond recovery.

Most designers assume components are mainly electronic parts. Unfortunately,

that is not true. A typical electronic product consists of three category of components;

1. Basic electronic parts like resistors, capacitors, transistors, ICs, etc
2. Commercial off-the-shelf (COTS) items like power supplies, display systems, keyboards, etc. These are electronic products which become part of the bigger product
3. Custom designed parts (typically mechani-

cal items) items like sheet-metal or plastic enclosures, cable harness, and printed circuit board assemblies (PCBAs). Typically, custom designed parts are specific to a product and are expensive as they have to be created through a fresh design.

Performance of a product depends on some key aspects like:

1. Cost of the components
2. Shape of the components (smaller parts need automated assembly and tend to have an expensive process)
3. Leadtime of the part's availability, that is, how much time the vendor will take to supply from the day purchase order is released. Longer the lead-time, larger the inventory companies have to maintain to keep the production line running
4. Life of the part (basically how long the component is available before the product production is stopped)
5. Quality of the COTS components and their availability
6. Components handling and storage. Most ICs use plastic encapsulation and tend to absorb moisture if not properly stored. The components have to be typically stored

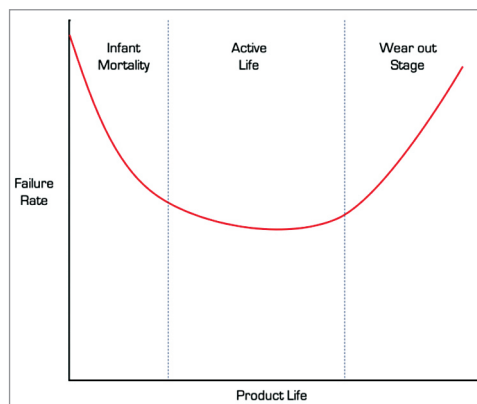


Fig. 1: Product failure chances during product's life